

USN

--	--	--	--	--	--	--	--	--	--

RV COLLEGE OF ENGINEERING®
 (An Autonomous Institution Affiliated to VTU)
 III Semester B. E. Examinations April-2024
 Common to AS, BT, CH, IEM, ME

STATISTICS, LAPLACE TRANSFORM AND NUMERICAL METHODS

Time: 03 Hours

Maximum Marks: 100

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.
3. Use of scientific calculator and hand book for mathematics is permitted.

PART-A

M CO BT

1	1.1	If the first four moments of a distribution about the value 25 of the variables are -3, 11, -29 and 71, then the second moment about the mean is _____.	02	2	1
	1.2	The equations of regression lines are $y = 0.5x + 1$ and $x = 0.4y + 3$. The correlation coefficient is _____.	02	2	2
	1.3	The real part of the complex function $f(z) = \log_e z$ in cartesian co-ordinate system is _____.	02	1	1
	1.4	Taylor's series expansion of $\cos z$ about the point $z = \frac{\pi}{4}$ is _____.	02	1	2
	1.5	The Laplace transform of the signal $f(t) = t + \cos 2t$ is _____.	02	1	1
	1.6	Region of convergence of $L[a^t]$ to hold good is _____.	02	1	1
	1.7	$L^{-1}\left[\frac{1}{4s-1}\right] =$ _____.	02	2	1
	1.8	If $F(s) = \frac{1}{s^2+4s+4}$, then the inverse Laplace transform of $F(s) =$ _____.	02	1	2
	1.9	The equation $u_{xx} + u_{yy} - 4u_x - 6u_y = 3$ is classified as _____.	02	1	1
	1.10	For the one dimensional wave equation $u_{xx} = u_{tt}$ with the boundary conditions $u(0, t) = 0, u(1, t) = 0, u_t(x, 0) = 0$ and $u(x, 0) = x - x^2$, the value of $u(0.25, 0.25)$ by taking $h = 0.25$ and $k = 0.25$ is _____.	02	2	2

PART-B

2

a

From the following frequency distribution, compute first three moments about the mean.

x	1	2	3	4	5	6
y	1	6	13	25	30	22

05

3

2

b

A simply supported beam carries a concentrated load P at its mid-point. Corresponding to various values of P , the maximum deflection Y is measured and is given in the following table.

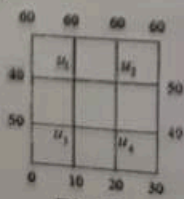
P	100	120	140	160	180	200
Y	0.45	0.55	0.60	0.70	0.80	0.85

Fit a law of the form $Y = a + bP$ and hence estimate Y when P is 150.

06

2

3

		Calculate the rank correlation coefficient from the following data:											
c		x	81	78	73	73	69	68	62	58	05	3	3
		y	10	12	18	18	18	22	20	24			
3	a	Determine the analytic function whose real part is $u = e^x(x \cos y - y \sin y)$ by Milne-Thomson method. Also find its harmonic conjugate v .									08	2	2
	b	Obtain the Laurent's series expansion for $f(z) = \frac{1}{(z+1)(z+3)}$ in the region											
		i) $1 < z < 3$ ii) $ z > 3$ iii) $ z < 1$									08	2	2
OR													
4	a	Find the analytic function $f(z) = u + iv$ where $u - v = 2xy + x^2 - y^2 + x - y$.									08	2	2
	b	Evaluate the integral $\int_C \frac{e^{2z}}{(z+1)(z-2)^2}$ where C is the circle $ z = 3$ using Cauchy's residue theorem.									08	2	2
5	a	A periodic function $f(t)$ of period $2a, a > 0$ is defined by $f(t) = \begin{cases} K & 0 \leq t \leq a \\ -K, & a < t \leq 2a \end{cases}$ Show that $L\{f(t)\} = \frac{K}{s} \tanh\left(\frac{as}{2}\right)$									08	2	2
	b	Obtain the Laplace transform of $\frac{e^{-t}-e^{-7t}}{t} + \left(2\sqrt{t} - \frac{3}{\sqrt{t}}\right)^3$									08	3	3
OR													
6	a	Express the following in terms of unit step function and hence find the Laplace transform of $f(t) = \begin{cases} 3t^2 & 0 < t < 3 \\ 5t & t > 3 \end{cases}$									08	2	2
	b	Find $L\left\{e^{-t} \left(\int_0^t t \sin 4t \, dt\right) + \cos 2t \sin 5t\right\}$									08	3	3
7	a	The signal in frequency domain is given by $\frac{s}{(s-3)(s^2+4)}$. Determine the transformation in time domain.									08	3	2
	b	Employing convolution theorem, find inverse Laplace transform of $\frac{1}{(s^2+a^2)^2}$									08	4	3
OR													
8	a	Evaluate $L^{-1}\left[\frac{s+1}{s^2-s+1}\right]$									08	3	2
	b	Solve the differential equation using Laplace transform given $\frac{d^2y}{dt^2} - 3\frac{dy}{dt} + 2y = 12e^{-t}$ when $y(0) = 2$ and $y'(0) = 6$.									08	4	3
9	a	Determine the value of $u(x, y)$ satisfying the Laplace equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ at the pivotal points of a square region with boundary values as shown in Fig 9a.											
													
		Carry out 3 iterations.									08	4	2
		Fig 9a											

b	Employ Bendre-Schmidth method with $h = 1$ to find the values of $u(x, t)$ upto $t = 1$ for the equation $2 \frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}$, $0 < x < 4, t > 0$ under the conditions $u(x, 0) = x(4 - x), \quad 0 < x < 4$ $u(0, t) = u(4, t) = 0, \quad t \geq 0$	08	3	3
OR				
10 a	The temperature distribution $u(x, t)$ is modeled as the one-dimensional heat equation $\frac{\partial u}{\partial t} = \frac{4}{5} \frac{\partial^2 u}{\partial x^2}$, $0 < x < 100, t > 0$ subject to the boundary conditions $u(0, t) = 0 = u(100, t)$, $t \geq 0$ and the initial condition $u(x, 0) = \begin{cases} x, & 0 \leq x \leq 50 \\ 100 - x, & 50 \leq x \leq 100 \end{cases}$ Determine the temperature distribution $u(x, t)$ by taking $h = 20$ and $k = 10$ using Schmidt relation for 5 levels.	08	3	2
b	Solve $\frac{\partial^2 u}{\partial t^2} = 16 \frac{\partial^2 u}{\partial x^2}$ taking $h = 1$ in five steps given that $u(x, 0) = x^2(5 - x)$, $u_t(x, 0) = 0$, $u(0, t) = u(5, t) = 0$	08	4	3